DAY 9 · CONSULTING RECOMMENDATION

Should Riverside Malls roll out 25 smart trolleys — and at what price?

Yes — at £200 per trolley per month.

£1,000 / month profit · 6-trolley margin of safety · break-even at 19 trolleys

Riverside Malls is exploring a smart-trolley pilot

SITUATION	Riverside Malls wants a tech-forward differentiator and is considering a fleet of 25 smart trolleys to give shoppers a frictionless cart and ops a shopper-flow signal.
COMPLICATION	Hardware + telematics + support is real money. A naive rollout could lose money every month — or worse, prove the unit economics don't work before the next budget cycle.
QUESTION	Should we go ahead with 25 trolleys, and at what price per trolley per month?

→ **ANSWER** Yes — at £200 per trolley per month. £1,000/month profit, 6-trolley margin of safety above break-even. Full case on slide 4.

Two numbers decide everything

What price per trolley per month turns a 25-trolley pilot into a profitable one?

REVENUE (per trolley / month)

Operator fee: P £

- Charged per active trolley, billed monthly.
- Only revenue counted in the base case.
- £1/visit shopper hire and ad slots are upside — called out, not booked.

COST (monthly)

v = £40 · F = £3,000 · T = 25

- v = variable cost per trolley (maintenance, telematics, support).
- F = fixed monthly cost (platform, depot, on-call).
- T = fleet size in the pilot = 25 trolleys.

THE FORMULA

Monthly profit = $T \times (P - v) - F$ **Break-even trolleys $n^* = F / (P - v)$**

Change one number, every cell updates. (See unit_economics.xlsx — Assumptions sheet.)

Run the pilot. Charge £200 per trolley per month.

MONTHLY PROFIT

£1,000

at £200 / trolley / month

BREAK-EVEN TROLLEYS

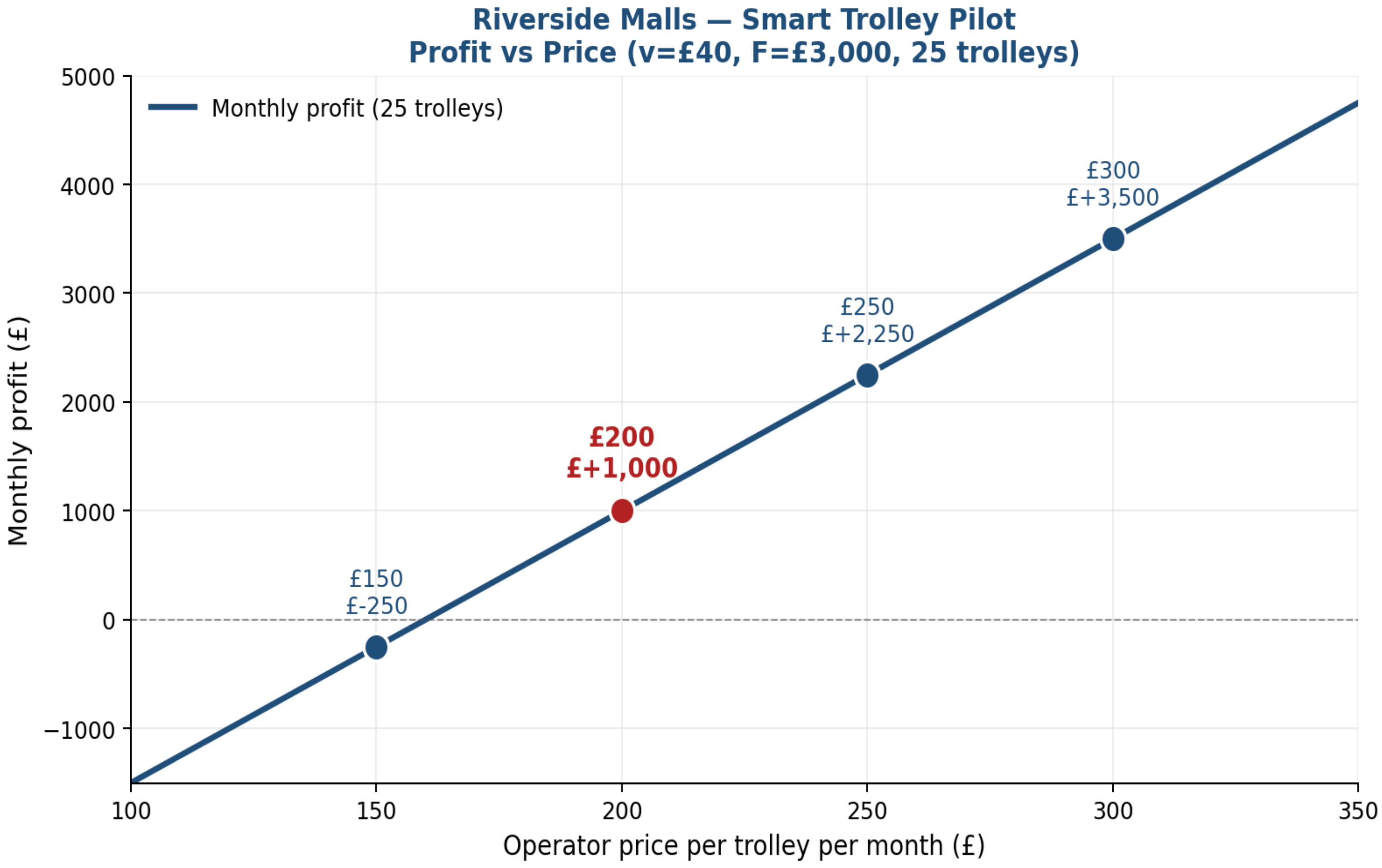
19

vs. 25 in the pilot → 6-trolley safety

WHY NOT £150?

At £150 the pilot loses £250/month — break-even needs 28 trolleys but we only have 25. Penetration pricing doesn't work until the fixed cost is amortised across more units.

£200 is the knee of the curve — and the only price with margin



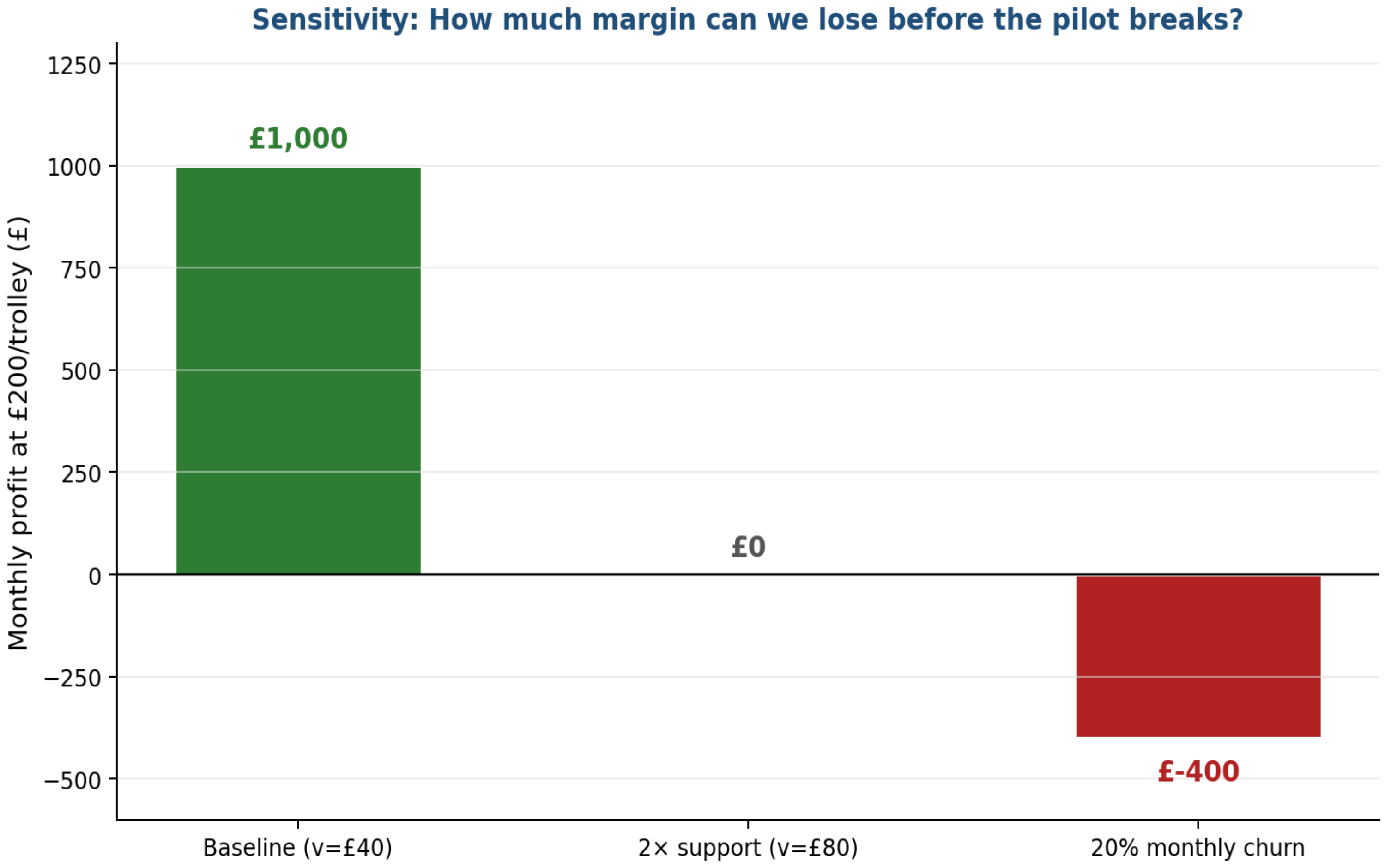
READ

The red dot is £200 — the first price at which the pilot turns green. Below it, every month is a loss. Above it, every extra £50/trolley adds another £1,250/month in profit.

GUARDRAIL

If price falls to £160 or below, profit goes negative. Anchor the sales conversation at £200 — don't negotiate on the headline number without a fleet-size concession.

The pilot survives 2× support cost — but not 20% monthly churn



Support cost doubles

We drop to exactly £0 profit. We must monitor cost-to-serve weekly in the first 90 days.

20% monthly trolley churn

Lose £400/month. A re-deployment + refurb capex plan is non-negotiable before we go live.

Trolley under-utilisation: if <19 trolleys are active, the pilot loses money. Recovery playbook + operator co-marketing required.

Three things to validate before we sign the contract

1 Confirm cost-to-serve

Shadow-quote support and telematics for 4 weeks with a real depot. The £40 variable cost is an estimate — moving it to £60 changes break-even from 19 to 22 trolleys.

2 Pilot a 5-trolley shadow fleet

Run a no-charge 4-week shadow deployment to measure actual utilisation and churn. Use the real data, not the model assumption of 100% utilisation.

3 Lock the £200 headline price

Set the operator-fee schedule at £200/trolley/month with a contractual volume clause: any sub-19-trolley month triggers a price-floor review.